Unsupervised Domain Adaptation for Semantic Segmentation of High-Resolution Remote Sensing Imagery Driven by Category-Certainty Attention

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***Abstract*— Semantic segmentation is an important task of analysis and understanding of high-resolution remote sensing images (HRSIs). The deep convolutional neural network (DCNN)-based model shows their excellent performance in remote sensing image semantic segmentation. Most of the existing HRSI semantic segmentation methods are only designed for a very limited data domain, that is, the training and test images are from the same dataset. The accuracy drops sharply once a model trained on a certain dataset is used for cross-domain prediction due to the difference in feature distribution of the dataset. To this end, this article proposes an unsupervised domain adaptation framework based on adversarial learning for HRSI semantic segmentation. This framework uses high-level feature alignment to narrow the difference between the source and target domains at the semantic level. It uses the category-certainty attention module to reduce the attention of the classifier on category-level aligned features and increase the attention on category-level unaligned features. Experimental results show that the proposed method performs favorably against the state-of-the-art methods in cross-domain segmentation.**

***Index Terms*— Category-certainty attention, domain adaptation, generative adversarial networks, semantic segmentation.**

I. Introduction

THE development of remote sensing technologies has made high-resolution remote sensing images (HRSIs) increasingly available. Semantic segmentation, which is an important task of HRSI analysis, aims to assign a specific semantic category to each pixel. It allows pixels with similar characteristics and attributes, such as color, intensity, and texture, to be classified into the same semantic category. Semantic segmentation using HRSIs plays an important role in many applications, such as urban traffic management and planning, precision agriculture, and disaster prediction. Many traditional methods [1]–[4] use artificially extracted features for pixel-by-pixel classification to characterize the content of remote sensing images. However, the design of manual features needs to rely on expert knowledge and experience. Moreover, the detailed feature information of remote sensing image brought by the increase in spatial resolution brings difficulty for the handcrafted feature to fully express the geo-object. Meanwhile, classification methods based on machine learning, such as support vector machine [5], random forest [6], and artificial neural network (ANN) [7], are widely used in the semantic segmentation of HRSIs. Although these classification models achieve satisfactory classification performance, they lack the ability of automatic extraction of more distinguishing high-level semantic features.

In recent years, deep convolutional neural networks (DCNNs) have shown outstanding performance in feature representation; thus, DCNN-based semantic segmentation methods, such as FCN [8], SegNet [9], UNet [10], PSPNet [11], and DeepLab [12], have been applied on HRSIs. Their results demonstrate that the semantic segmentation field has achieved significant progress to date. Moreover, many novel semantic segmentation methods have been developed based on these baseline models. For example, an object-based convolutional neural network [13] is proposed for urban land use classification. The multiscale adversarial features are shared into a U-shaped semantic segmentation model for HRSI semantic segmentation [14]. SPMF-Net [15] is a weakly supervised semantic segmentation method that uses the image-level label as supervision information for building detection. However, all these advanced methods tend to be sensitive to the variation in the data domain. If a domain gap exists between the train and test images, then the performance of the methods falls dramatically due to the domain bias between the source and target datasets. The appearance characteristics, such as color, texture, spatial resolution, and scene context, vary because of the difference in monitoring aims, remote sensing sensors, observational weather, and solar illumination, as shown in Fig. 1. That is, knowledge transfer performance from the source domain to target domain inevitably needs to be considered due to the domain shift between different HRSI datasets.

Domain adaptation is proposed to meet the domain shift between source and target datasets for addressing the above-mentioned issue. Domain adaptation, which is a branch of transfer learning, executes new tasks on the unlabeled target domain with the knowledge learned from the labeled source domain [16]. Early domain adaptation works have mainly focused on image classification tasks. Numerous DCNN-based methods [17]–[21] are developed to learn domain-invariant representations through minimizing the domain distribution

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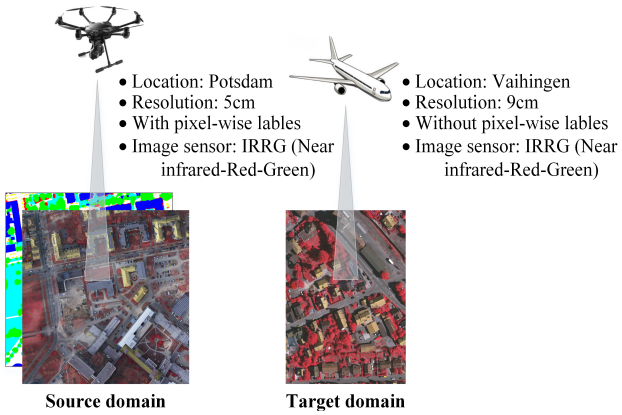

Fig. 1. Cross-domain semantic segmentation in aerial imagery.

discrepancy between source and target images. Among these domain adaptation methods, maximum mean discrepancy (MMD) [22] is a common distance metric to minimize feature distributions between the source and target domains. Subsequently, some extended statistics, such as multikernel MMD and domain-confusion loss [17], [19], [20], are explored to match the domain discrepancy. Adversarial learning [23] is another popular choice for domain adaptation methods; it reduces the domain shift through a minimax game between the generator and the discriminator. Adversarial discriminative domain adaptation [21] originally introduces adversarial loss on the high-level features of the two domains and achieves better classification performance than statistical metric methods on the target domain dataset. Shen *et al.* [24] proposed Wasserstein distance-guided adversarial learning to learn domain invariant feature representations. The adversarial learning-based work [25] uses multiple discriminators (MADA) to solve the mode collapse problem in the domain adaptation.

In recent years, some domain adaptation methods have been proposed for semantic segmentation tasks. The first domain adaptation work of semantic segmentation [26] aligns two domains in the feature space from the global and local levels. Curriculum domain adaptation [27] learns global label distributions of images and local label distributions of landmark superpixels to minimize the domain gap in semantic segmentation. MCD_DA [28] attempted to utilize a task-specific classifier and a feature generator to align the distribution of source and target datasets. ADVENT [29] addresses the task of unsupervised cross-domain semantic segmentation with the loss based on the entropy of the pixel-wise predictions. A joint global and class-wise domain adversarial learning framework [30] is proposed to conduct the classification of cross-city road scenes. AdaptSegNet [31] aligns the output space by using a multilevel adversarial network, which leads to more effective performance in semantic segmentation. Zhang *et al.* [32] proposed a robust class-wise matching scheme by attempting to simultaneously minimize the distribution discrepancy between the same classes and maximize the distribution discrepancy between different classes across domains. The Transferrable Prototypical Networks [33] reduced the domain discrepancy both at class-level and sample-level. Luo *et al.* [34] used a category-level adversarial network (CLAN) to enhance local semantic consistency during the trend of global alignment. The work in [35] uses different strategies to reduce domain drift for foreground and background classes under the different effects of domain drift on the foreground and background. All the above-mentioned domain adaptation methods pursue the alignment of the source and target domains at the pixel level.

Driven by the work of image-to-image translation, another domain adaptation method aligns two domains at the pixel and feature levels. It usually consists of two separated subnetworks: an image-to-image translation subnetwork and a semantic segmentation adaptation subnetwork. Before training the semantic segmentation model, image translation technology is used to map the source domain image to the target domain for reducing the discrepancy between domains. CyCADA [36] uses cycle-consistency constraint to encourage the cross-domain transformation before training the segmentation models. DCAN [37] translates source domain images to target domain in an appearance before feature-aligning in feature level. A bidirectional learning framework [38] is introduced to motivate the image translation and segmentation adaptation models for promoting each other alternatively. As a result, the domain gap reduces gradually.

Although these aforementioned domain-adaptive methods have introduced impressive performance in cross-domain semantic segmentation, they are developed for natural image datasets. The effect is very unsatisfactory when these methods are directly used for HRSI segmentation because natural images are quite different from HRSI in various aspects, such as shooting angle, space complexity, and image resolution. To meet this challenge, Benjdira *et al.* [39] developed a GAN-based algorithm of cross-domain semantic segmentation for aerial imagery (Benjdira's). This algorithm first translates the source domain image to the target domain image using a GAN model. Then, it fine-tunes the semantic model that has been well-trained on original source datasets. However, the performance of semantic segmentation is limited by the quality of image-to-image translation. The accuracy of semantic segmentation is reduced once the translation fails. In addition, the image-to-image translation can only make the source domain images similar to the target domain images in terms of image style (such as color distribution and texture features), and the difference in image context information and class representation is difficult to reduce. Notably, most of the abovementioned methods consider the entire image only with the same attention when bridging the domain gap, that is, the local difference is ignored. In fact, domain knowledge transfer for HRSIs will lead to different degrees of difficulty when faced with different regions and categories in one image due to the effects of spatial resolution, appearance distribution, object size, and contextual information of the scene. For example, after a period of domain adaptation training, some regions and categories will not be well adapted, while some other regions and categories have already been adapted. In this case, the former should be paid more attention to classification. By contrast, the latter should be paid less attention during

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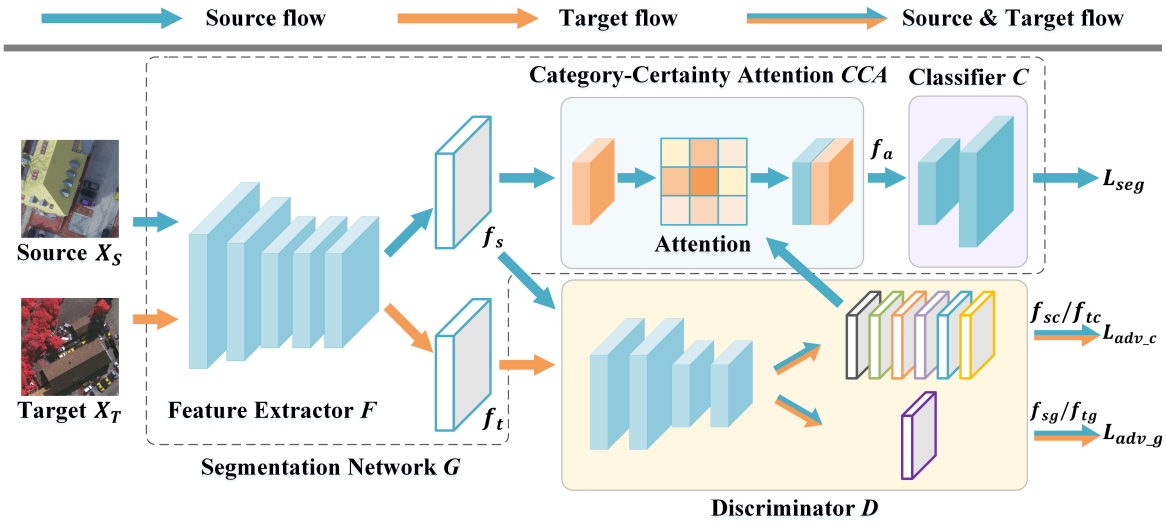

Fig. 2. Overview of the proposed model.

classification; otherwise, the negative transfer may occur. Therefore, the abovementioned methods may be inappropriate for the cross-domain semantic segmentation of HRSIs.

Attention mechanism (AM) has played an important role in deep neural networks and is widely used in machine translation [43], object detection [44], image caption [45], [46], and speech recognition [47]. AM can help the network focus on the important information and ignore the irrelevant information. In recent domain adaptation works, an attention alignment strategy was proposed in [48] to transfer knowledge in all the convolutional layers, or the domain adaptation problem was solved by strengthening the attention of the classifier on adaptable regions [49]. Although AMs have achieved well performance in domain adaptation of scene classification, it is still a big challenge in domain adaptation of HRSI semantic segmentation. In HRSI, geo-objects are of various sizes, scene background are complex, and geo-objects have a lot of detailed information. These AMs cannot achieve satisfactory results when they are directly applied to the domain adaptation of HRSI semantic segmentation.

On the basis of the abovementioned analyses, in this article, we propose an unsupervised domain adaptation semantic segmentation model (UDAS) driven by category-certainty attention to handling the cross-domain semantic gap in HRSIs. Our framework consists of two major modules: segmentation network G and global & category-level discriminator D. Meanwhile, the segmentation network consists of a feature extractor, a category-certainty attention module, and a classifier. In our method, global and the category-level discriminator can enhance the local semantic consistency while ensuring global consistency. Furthermore, the category-certainty attention module can enhance the attention of the classifier to unaligned regions and reduce the attention to well-aligned regions to enhance the performance of the classifier on the target domain. The main contributions of the proposed network can be summarized as follows.

1) We propose an unsupervised domain adaptation method for HRSI semantic segmentation by emphasizing the importance of category-level alignment.
2) A category-certainty attention module is developed to improve the performance of the classifier on the target dataset by focusing the training of the segmentation model on the misaligned regions in the category-level.
3) The category-level adversarial loss can effectively improve the category-level feature alignment.

The remainder of this article is organized as follows. Section II explains the proposed method. Section III presents the experimental details and results. Section IV concludes our work.

II. UDAS Method

A. Problem Description

We focus the problem of unsupervised domain adaptation in the semantic segmentation of HRSIs. We use the data X_S with pixel-level labels Y_S as source domain $D_S \in P_S$ and the data X_T without labels as target domain $D_T \in P_T$, where P_S and P_T are the distributions of source and target domains, respectively. The goal is to train a semantic segmentation model that can correctly make a pixel-level prediction for the target domain by transferring knowledge learned from the source domain. To deal with this task, we propose an unsupervised domain adaptation framework for pixel-level classification of HRSIs.

B. Model Architecture

Our proposed model architecture is illustrated in Fig. 2. The framework consists of two modules: the segmentation network G and the global and category-level discriminator D. Meanwhile, the segmentation network contains a feature extractor F, a category-certainty attention module CCA, and a classifier C. The feature extractor aims to extract features separately and establish a common invariant feature space for source and target images inspired by adversarial learning. The classifier aims to perform precise pixel-level classification for source and target images. For achieving high-quality segmentation results, we adopt the DeepLab-v2 [12] framework with ResNet-101 [40] model pre-trained on the source dataset as our

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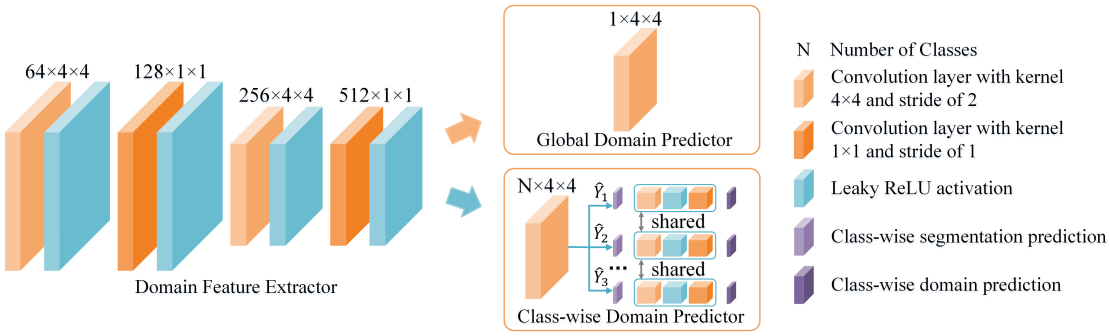

Fig. 3. Architecture of global- and category-level discriminator.

backbone of feature extractor and classifier. We also apply an up-sampling layer along with the softmax output to match the size of the input image. For better domain adaptation performance, we insert a category-certainty attention module after feature extractor F (stage-4 of the ResNet-101). This module is designed to draw category-level attention to the regions that are not well adapted and reduce the attention to the regions that have already been aligned during classification. The global and category-level discriminator attempts to distinguish whether the source and target features come from the same domain for enabling learning of domain global and category invariant feature representation.

We first train the segmentation network G using source images X_S with corresponding semantic labels Y_S and target images X_T without ground truth semantic labels. Specifically, given a source domain image $x_s \in X_S$, feature extractor F outputs a feature map f_s, which is input to global and category-level discriminator D to yield the global-certainty map f_{sg} and the category-certainty map f_{sc}. The category-certainty attention module CCA receives f_s and f_{sc} as input and outputs the features f_a with attention. Finally, f_a is forwarded to the classifier C to yield the pixel-by-pixel classification, which is used to calculate a segmentation loss to optimize the G. We also use a target domain image $x_t \in X_T$ as the input to the feature extractor F to obtain a target feature map f_t, and input it to D to generate an adversarial loss to optimize the feature extractor F.

We then optimize D with the source feature map f_s and target feature map f_t. Our goal is to make the feature difference between the source and target domains as small as possible. Thus, we forward the two features to the global and category-level discriminator D to distinguish whether the input is from the source or target domain at the global level and category level.

1) Global and Category-Level Discriminator: Most domain adaptation methods pursue the global distribution alignment while ignoring the local joint distribution shift, which will lead to negative transfer. This condition results in the false alignment of two originally well-aligned distributions across domains. Therefore, a domain adaptation method with global and local semantic consistencies should be explored. Recent works [34], [41] introduce a CLAN through a cotraining strategy to enforce local category-level consistency during the process of global alignment. However, the cotraining strategy used in these methods causes a long-time consumption in the process of parameter updating for the deep neural network.

On the basis of the abovementioned analyses, we propose a novel global and category-level discriminator D to promote the category-level feature distribution alignment while aligning the global-level feature distribution. As shown in Fig. 3, the global and category-level discriminator consists of three modules: a domain feature extractor, a global domain predictor, and a class-wise domain predictor. The domain feature extractor uses an architecture of four convolution layers, each of which is followed by a leaky ReLU [42] parameterized by 0.2. The channel number of the four convolution layers are [64, 128, 256, 512], respectively. The global domain predictor outputs the global domain labels, it uses a 4×4 convolution layer with a channel number of 1. The class-wise domain predictor outputs the class-wise domain labels, it first uses a 4×4 convolution layer with the channel number of N to obtain a segmentation prediction $\hat{Y}$, which is supervised by the ground truth of source data, where N is equal to the number of classes. Each class-wise segmentation output $\hat{Y}_i (i = [1, 2, \ldots, N]) \in \hat{Y}$ contains rich spatially and locally information, although the images from the source domain and the target domain are very different in appearance, their class-wise segmentation outputs have great similarities in spatial layout and local context with other classes. Discriminating the domain information of each class-wise segmentation prediction obtained by source and target domain can encourage the alignment of the two domains in the category. Based on the above analysis, we then input each class-wise segmentation prediction $\hat{Y}_i (i = [1, 2, \ldots, N])$ to an architecture of 4×4 convolution—leaky ReLU parameterized by 0.2—1×1 convolution layer to obtain the class-wise domain predictions. Each class-wise domain prediction forms the category-certainty feature map A.

2) Category-Certainty Attention: The discriminator is used to distinguish whether the input is from the source or target domain, and its output represents the probability that the input comes from a different domain. The probability with a value of 0 means the input comes from the source domain, while the probability with a value of 1 represents that the input comes from the target domain. The output probability is equal to 0.5 when the discriminator cannot determine whether the input is from the source or target domain. Therefore, the certainty estimation of the discriminator can indicate the

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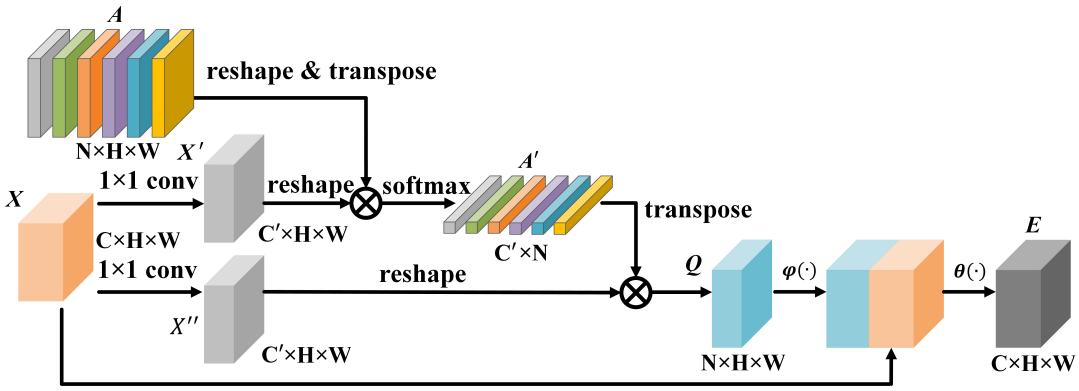

Fig. 4. Architecture of category-certainty attention module.

poorly aligned and well-aligned regions. The regions that have been already adapted will confuse the discriminator for predicting the domain label. Thus, the discriminator will be ambiguous in these regions with a probability of 0.5. On the contrary, the discriminator can determine in the poorly aligned regions with the probability of 0 or 1. Consequently, if the classifier pays more attention to these poorly aligned regions during training, then better classification performance will be achieved.

We extend the AM to the domain adaptation of HRSI semantic segmentation in this study. Different from previous works, we propose an attention module based on category-certainty obtained by global and category-level discriminators to increase the attention of classifiers on category-level poorly aligned features while reducing the attention of classifiers on those already aligned.

As shown in Fig. 4, given a feature $X \in R^{C\times H\times W}$, which is output from the convolution layer after stage-4 of ResNet-101 in our implementation, the category-certainty attention module first applies a 1×1 convolutional layer to generate a feature map $X' \in R^{C'\times H\times W}$, where C' is the reduced channel number of the feature X for less computational cost. In this work, we set C' to $C/8$, where C is the channel dimension of feature X. Then, we reshape it to $R^{C'\times M}$, where $M = H \times W$ is the number of pixels. Thereafter, we perform a matrix multiplication between the reshaped X' and the category-certainty feature map $A \in R^{N\times H\times W}$ from the discriminator (N is the number of categories). We also apply a softmax layer to calculate the category-certainty affinity attention map $A' \in R^{C'\times N}$. The category-certainty affinity operation is defined as follows:

$$a'_{j,i} = \frac{\exp\left(X'_j \cdot A_i\right)}{\sum_{i=1}^{N}\exp\left(X'_j \cdot A_i\right)} \tag{1}$$

where $a'_{j,i} \in A'$ measures the correlation between the certainty of the ith category of feature A and jth channel of feature X', $j = [1, 2, \ldots, C'], i = [1, 2, \ldots, N]$.

Meanwhile, we feed feature X into a convolution layer to generate a new feature map $X'' \in R^{C'\times H\times W}$ and reshape it to $R^{C'\times M}$, where $M = H \times W$. Then, we perform a matrix multiplication between the transposed attention map A' and the reshaped feature X'' and reshape it to the $Q = R^{N\times H\times W}$. Finally, we concatenate the original local feature X and the attention map Q to obtain the final output $E = R^{C\times H\times W}$ as follows:

$$E = \theta(\text{cat}(\varphi(Q), X)) \tag{2}$$

where $\varphi(\cdot)$ and $\theta(\cdot)$ denote transformation functions implemented by 1×1 conv $\rightarrow$ BN $\rightarrow$ ReLU. Thus, the feature E at each position is a weighted value of the category-certainty feature and original features. Therefore, this feature can enhance the classifier to selectively pay attention to the regions that are well aligned and misaligned according to the affinity attention map. Accordingly, the classification performance of the classifier on the target domain will be improved.

C. Network Training

1) Segmentation Model Training: In unsupervised domain adaptation, since the ground truth label is only available in the source domain, we train the segmentation network G with source images in a fully supervised way. The semantic segmentation loss is defined as

$$L_{\text{seg}}(X_S, Y_S) = -E_{(x_s,y_s)\sim(X_S,Y_S)}\sum_{i=1}^{N} y_s^{(i)}\log G(x_s)^{(i)} \tag{3}$$

where $L_{\text{seg}}(X_S, Y_S)$ is the cross-entropy loss using ground truth annotations in the source domain, G represents the segmentation network, N represents the number of classes, $x_s \in X_S$ represents the images from the source domain, and y_s represents the ground truth from the source domain.

And when the target image is given, we calculate the adversarial loss using a discriminator. It should be noted that this loss is only used to optimize the feature extractor F

$$L_{\text{adv}}(X_T) = \lambda_{\text{adv_}g}L_{\text{adv_}g}(X_T) + \lambda_{\text{adv_}c}L_{\text{adv_}c}(X_T) \tag{4}$$

where $L_{\text{adv_}g}(X_T)$ donate the global adversarial loss on target data, and $L_{\text{adv_}c}(X_T)$ donate the category-level adversarial loss on target data. $\lambda_{\text{adv__}g}$ and $\lambda_{\text{adv_}c}$ are weights used to balance these two losses. The $L_{\text{adv_}g}(X_T)$ and $L_{\text{adv_}c}(X_T)$ can be defined as follows, respectively:

$$L_{\text{adv_}g}(X_T) = -E_{x\sim P_T(x)}\left[\log D_g(F(X_T))\right] \tag{5}$$

$$L_{\text{adv_}c}(X_T) = -E_{x\sim P_T(x)}\left[\sum_{i=1}^{N}\log D_c^i(F(X_T))\right] \tag{6}$$

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where $P_T(x)$ denotes the data distribution of target domain; $F(X_T)$ denote the target features from feature extractor F; $D_g(x)$ denotes the out of global-level discriminator, $D_c(x)$ denotes the out of category-level discriminator; and N represents the number of categories.

Therefore, the total loss function for the segmentation network is defined as

$$L(X_S, X_T) = L_{\text{seg}}(X_S, Y_S) + \lambda_{\text{adv}} L_{\text{adv}}(X_T). \tag{7}$$

2) Discriminator Training: The discriminator takes source and target features and aligns its distributions at the global level and category level. Its loss is defined as follows:

$$\begin{aligned} &L_{\text{adv}}(X_S, Y_S, X_T) \\ &\quad = \lambda_{\text{adv_}g} L_{\text{adv_}g}(X_S, X_T) \\ &\qquad + \lambda_{\text{adv_}c} L_{\text{adv_}c}(X_S, X_T) + \lambda_{\text{seg}} L_{\text{seg}}(X_S, Y_S) \end{aligned} \tag{8}$$

where $L_{\text{adv_}g}(X_S, X_T)$ denotes the global-level adversarial loss, which aligns features of source and target at the global level. $L_{\text{adv_}c}(X_S, X_T)$ denotes the category-level adversarial loss, which aligns features of source and target at the category level. And $L_{\text{seg}}(X_S, Y_S)$ denotes the semantic segmentation loss that supervises the first convolution layer of class-wise domain predictor to obtain semantic prediction $\hat{Y}$ for the further prediction of class-wise domain information. $\lambda_{\text{adv_}g}$, $\lambda_{\text{adv_}c}$ and λ_{seg} are weights used to balance these three losses. In this work, we set $\lambda_{\text{adv_}g}$, $\lambda_{\text{adv_}c}$ and λ_{seg} to 1.0, 1.0, 0.5, respectively. The semantic segmentation loss is defined as formula 4, the $L_{\text{adv_}g}(X_S, X_T)$ and $L_{\text{adv_}c}(X_S, X_T)$ are defined as follows, respectively:

$$\begin{aligned} L_{\text{adv_}g}(X_S, X_T) = &-E_{x\sim P_s(x)}\left[\log D_g(F(X_S))\right] \\ &- E_{x\sim P_t(x)}\left[\log\left(1 - D_g(F(X_T))\right)\right] \end{aligned} \tag{9}$$

$$\begin{aligned} L_{\text{adv_}c}(X_S, X_T) = &-E_{x\sim P_s(x)}\left[\sum_{i=1}^{N} \log D_c^i(F(X_S))\right] \\ &- E_{x\sim P_T(x)}\left[\sum_{i=1}^{N} \log\left(1 - D_c^i(F(X_T))\right)\right] \end{aligned} \tag{10}$$

where N represents the number of categories; $P_s(x)$ denotes the data distribution of source domain; $P_T(x)$ denotes the data distribution of target domain; $F(X_S)$ and $F(X_T)$ denote the source and target features from feature extractor F, respectively; $D_g(x)$ and $D_c(x)$ denote the out of global-level discriminator and category-level discriminator, respectively.

To train the proposed network, we alternately train the segmentation network G and discriminator D in one iteration. In each training batch, we firstly forward the source images X_S and target images X_T to optimize the segmentation network G for $L(X_S, X_T)$ in formula (7) and obtain the source features f_s and target features f_t. Then we pass them together to the discriminator D for optimizing $L_{\text{adv}}(X_S, Y_S, X_T)$ in formula (8).

D. Inference

For inference, we input the target image to F to output a feature map X, and input it to global & category-level discriminator D to yield the global-certainty map and the category-certainty map A. Since the value of 0 in A indicates input comes from the source domain and the value of 1 indicates input comes from the target domain, and we want the classifier to pay more attention to the region identified as the source domain (that is, the region with the value close to 0) on the target domain image, we set $A = 1 - A$. Then the category-certainty attention module CCA receives X and A as input and outputs the features E with attention. Finally, E is forwarded to the classifier C to yield the pixel-by-pixel classification.

III. Experiments and Result Analysis

A. Data Description

To validate our methodology, we conduct our experiments on three very high-resolution (VHR) datasets: Potsdam dataset [50], Vaihingen dataset [51], and Gaofen-2 dataset [52] with two adaptation tasks, e.g., Potsdam → Vaihingen and Potsdam → Gaofen-2. Among them, Potsdam and Vaihingen datasets provided by the International Society for Photogrammetry and Remote Sensing (ISPRS), and Gaofen-2 dataset provided by the Zhongke Xingtu cup. Given that the Potsdam dataset is larger than the Vaihingen and Gaofen-2 dataset, the Potsdam dataset is taken as the source domain and the Vaihingen and Gaofen-2 dataset are taken as target domains.

1) Potsdam Dataset: Potsdam dataset covers a typical historical city with a dense settlement structure. It consists of 38 VHR true orthophoto tiles (TOPs) with a size of 6000 × 6000 pixels and a spatial resolution of 5 cm. The TOPs are TIFF files with different channel compositions: IR-R-G, R-G-B, and IR-R-G-B, where IR, R, G, and B stand for near-infrared, red, green, and blue bands. A total of 24 TOPs among them have been manually pixel-wise labeled into six land cover classes, namely, impervious surfaces, buildings, low vegetation, tree, car, and clutter, as shown in Fig. 5. We choose the 24 labeled images with ground truth labels as the source domain for training.

2) Vaihingen Dataset: Vaihingen dataset covers a small village with detached buildings. It consists of 33 TOPs with size near 2000 × 2000 pixels and a spatial resolution of 9 cm, which contain the same six classes as the Potsdam dataset. The TOPs of Vaihingen have compositions of IR-R-G, where IR, R, and G stand for near-infrared, red, and green bands. We choose 26 TOPs as target datasets for training, and seven other TOPs are used for testing.

3) Gaofen-2 Dataset: Gaofen-2 dataset captured by Gaofen-2 (GF-2) consists of 500 images with a size 512 × 512 pixels and a spatial resolution of higher than 1 m. Each image is pixel-wise marked with nine land cover classes, namely, road, buildings, low vegetation, tree, land, water, vehicle, impervious surfaces, and others. To transfer from Potsdam → Gaofen-2, we adjust the category of gaofen-2 dataset to the same as the Potsdam dataset as shown in Fig. 6(c). We choose 350 images as target datasets for training, and 150 other images are used for testing.

Fig. 7 represents the proportions of each category pixels to the total number of pixels of Potsdam, Vaihingen, and

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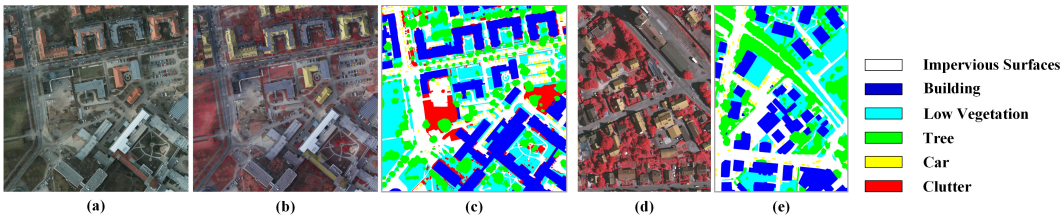

Fig. 5. Samples of Potsdam and Vaihingen datasets. (a) RGB image. (b) and (d) IRRG images. (c) and (e) Labeled ground truth.

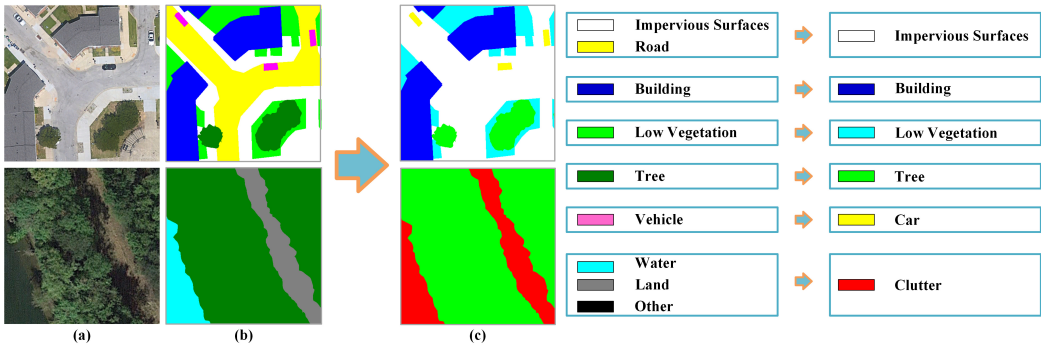

Fig. 6. Samples of Gaofen-2 dataset. (a) RGB images. (b) Original labeled ground truth. (c) Adjusted labeled ground truth.

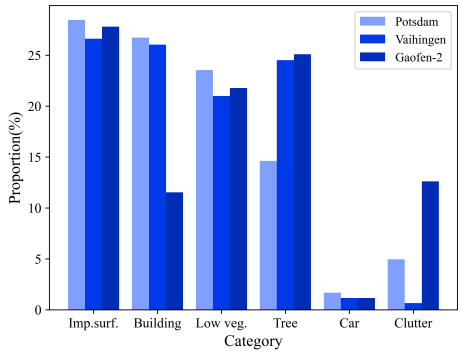

Fig. 7. Pixel proportions of different categories of Potsdam, Vaihingen, and Gaofen-2 datasets.

Gaofen-2 datasets. As shown in this figure, the different geo-object categories are proportionally unbalanced. For example, the pixel number of cars and clutter/background is far less than others.

The domain gaps between Potsdam & Vaihingen and Potsdam & Gaofen-2 datasets are mainly manifested in the factors of appearance characteristics, image resolution, and class representation. In principle, differences in appearance characteristics can be eliminated by the alignment of low-level features, and differences in class representation can be eliminated by the alignment of high-level features. However, image resolution usually involves objects of different scales and scenes in different contexts, and the alignment of features and pixels has difficulty eliminating the negative effect of resolution differences on knowledge transfer. As shown in Fig. 8, compared with Potsdam images, Vaihingen and Gaofen-2's images contain smaller-scale objects, larger spatial scene layout, and more spatial context information at the same size patch of 512 × 512 due to the difference of resolution. This condition will create an insurmountable semantic gap for cross-domain semantic segmentation of HRSIs. We crop the Potsdam and Vaihingen TOPs into patches with sizes of 960 × 960 and 512 × 512, respectively, according to their resolution ratio of 5:9, and keep the Gaofen-2's original image size of 512 × 512. This way eliminates the effects of resolution difference on domain adaptation while ensuring that the information of the image is not compressed. Accordingly, the three datasets have objects with similar scale and spatial context information. Fig. 8 shows some samples from Potsdam, Vaihingen, and Gaofen-2 images.

B. Experimental Settings and Evaluation Metrics

All the networks implemented in the experiments of this study are trained on the PyTorch framework. During the training, the stochastic gradient descent (SGD) [42] optimizer with a momentum of 0.9 and the weight decay of 10^{-4} are used to train the segmentation networks, while the Adam optimizer [53] with the momentum of 0.9 and 0.99 is used for training the discriminator. For SGD, the initial learning rate is set as 2.5×10^{-4} and decayed by a polynomial learning rate with a power of 0.9. For Adam, the learning rate is initialized to 10^{-4} and decayed using the same polynomial learning rate with a power of 0.9. We train the network for a total of 150k iterations.

Four common evaluation metrics are used, namely, F_1-score, OA, MA, and mIoU [8], to evaluate the performance of different methods for semantic image segmentation. Higher values of these metrics correspond to better performance of segmentation.

F_1-score is an evaluation metric of multiple classification problems. It can be calculated as follows:

$$F_1 = 2 \times \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \tag{11}$$

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TABLE I
RESULT EVALUATION OF COMPARATIVE EXPERIMENTS ON THE ADAPTATION TASK POTSDAM → VAIHINGEN

Methods	Imp. surf.	Building	Low veg.	Tree	Car	Clutter	OA	MA	mIoU
Baseline	52.12	59.38	44.36	60.69	2.74	8.92	48.88	48.18	25.93
MCD_DA [28]	69.26	49.35	53.45	61.22	6.62	12.48	56.28	49.20	29.40
ADVENT [29]	67.26	69.61	50.32	61.74	1.37	20.38	59.88	53.72	32.39
Benjdira's [39]	63.99	71.06	52.38	65.19	22.32	13.62	63.94	53.70	35.80
AdaptSegNet [31]	75.51	84.21	60.42	67.74	24.54	24.93	71.01	61.22	42.69
CLAN [34]	75.58	83.63	62.01	**67.76**	20.85	25.32	71.10	60.41	42.49
Ours	**80.30**	**84.24**	**65.59**	64.18	**40.57**	**28.85**	**73.62**	**63.03**	**45.91**

Fig. 8. Potsdam, Vaihingen, and Gaofen-2 image patches with different resolutions. For visual effects, we resize the 960 × 960 Potsdam image patches to 512 × 512, but we still use 960 × 960 image patches in actual training.

where Precision is the true positive (TP) pixels divided by the sum of TP and false positive (FP) pixels;

It is defined as follows:

$$\text{Precision} = \text{TP}/(\text{TP} + \text{FP}). \tag{12}$$

The Recall is the ratio of TP pixels and the sum of TP and false-negative (FN) pixels; it is defined as follows:

$$\text{Recall} = \text{TP}/(\text{TP} + \text{FN}). \tag{13}$$

OA measures the overall accuracy of semantic segmentation, MA measures the mean accuracy of semantic segmentation, and mIoU (mean Intersection over Union) calculates the Intersection over Union on each category. If we define n_{ij} is the number of pixels of class i predicted to class j, n_{ji} is the number of pixels of class j predicted to class i, N is the number of the classes, then OA, MA, and mIoU are defined as follows:

$$\text{OA} = \sum_i n_{ii} \Big/ \sum_i \sum_j n_{ij} \tag{14}$$

$$\text{MA} = (1/N) \sum_i \left(n_{ii} \Big/ \sum_j n_{ij} \right) \tag{15}$$

$$\text{mIoU} = (1/N) \sum_i n_{ii} \Big/ \left(\sum_j n_{ij} + \sum_j n_{ji} - n_{ii} \right). \tag{16}$$

C. Experimental Result Analysis

1) Comparative Studies on Potsdam → Vaihingen: We conduct comparative experiments on task Potsdam (IRRG) → Vaihingen (IRRG) with the state-of-the-art domain adaptation methods [28], [29], [31], [34], [39] listed in Table I to verify the performance of the proposed domain adaptation method for semantic segmentation of HRSIs. For fairness, the backbones of feature extractors in all comparison methods use the ResNet-101 model pre-trained on the source dataset. Table I shows that all the domain adaptation methods achieve improved performance compared with the baseline that only utilizes the source domain for training and then directly tests on the target domain. This result shows that the strategy of domain adaptation can bridge the distribution shift of different domains to some extent. Notably, the proposed method achieves the highest OA, MA, and mIoU values of 73.62%, 63.03%, and 45.91%, respectively, compared with other state-of-the-art methods. Among all the comparison models, the model with the worst performance is MCD_DA, which has OA, MA, and mIOU of 56.28%, 47.20%, and 29.40%, respectively. Compared with MCD_DA, the proposed method has the improvement of 17.34% on OA, 13.83% on MA, and 16.51% on mIoU. The model with the best accuracy among all comparison models is CLAN, which has OA, MA, and mIOU of 71.10%, 60.41%, and 42.49%, respectively. Compared with CLAN, the proposed method increases by 2.52%, 2.62%, and 3.42% on OA, MA, and mIoU, respectively. These comparative advantages demonstrate that the proposed method can achieve excellent cross-domain semantic segmentation performance on Potsdam → Vaihingen.

In Table I, the F_1 score of our method in the categories of Impervious Surfaces, Building, Low Vegetation, Car, and Clutter reaches the best accuracies of 80.30%, 84.24%, 65.59%, 40.57%, and 28.85%, respectively, compared with those of other models. Compared with the second highest

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score performance, our method is improved by 4.72%, 0.03%, 3.58%, 16.03%, and 3.92%, respectively. This result denotes the effectiveness of category-level alignment. The reason for the less improvement in F_1 score on Building is that the buildings in the Potsdam and Vaihingen datasets have high similarity in shape, texture, and class representation, which has a relatively small effect on knowledge transfer. Therefore, they can achieve a good transfer performance without applying the global & category-level adversarial loss and category-certainty attention module used in our method.

As shown in Fig. 9, our method has a great improvement in F_1 score in each category compared with the baseline trained on source images without adaptation. For the Car category, the improvement is the most remarkable, which gains an increase of 16.03% compared with the second highest improvement. The reason is that the size of the car is small, the color difference is large, and the car is always occluded by the tree, which leads to the difficulty of domain alignment. This difficulty greatly affects the performance of other domain adaptation methods in the car category. However, the proposed method can adaptively add attention to unaligned categories and regions according to the alignment degree of each category. This way improves the performance of the classifier for categories and regions that is hard to adapt.

Table I also shows that our method does not reach the highest F_1 score in the Tree category. The reason is that, unlike those in the Vaihingen dataset, the trees in the Potsdam dataset are sparser and do not completely cover the car and building, which makes knowledge transfer of trees more complex in cross-domain semantic segmentation. CLAN and AdaptSegNet align two domains by considering the spatial layout information and providing a certain prior knowledge for the domain adaptation of the Tree category. Thus, they can obtain better domain adaptation segmentation results. Nevertheless, Fig. 9 shows that our method still achieves a 3.49% improvement compared with the baseline Deeplab-v2 (ResNet-101) trained on source images without adaptation. In short, the proposed method can improve the classification performance of the classifier on the target domain by adaptively adding attention to unaligned regions and categories according to the degree of domain alignment.

Fig. 10 shows the qualitative analysis results from Potsdam $\rightarrow$ Vaihingen using our method and comparison methods. In the figure, the white, blue, cyan, green, yellow, and red regions indicate the categories of impervious surfaces, buildings, low vegetation, trees, cars, and clutter/background, respectively. As shown in Fig. 10, MCD_DA and ADVENT use adversarial learning methods to make the source and target domain features as the same as possible. Accordingly, MCD_DA uses a task-specific decision boundary method to further reduce the distribution difference of two domain features. ADVENT reduces the distribution difference by minimizing the entropy of the prediction result of the target domain image. MCD_DA and ADVENT obtain unsatisfactory testing results, as shown in Fig. 10(a) and (b), due to the lack of alignment of local distributions. Benjdira's methods use unpaired pixel-to-pixel generative adversarial networks to translate source images to target images. Then, they fine-tune

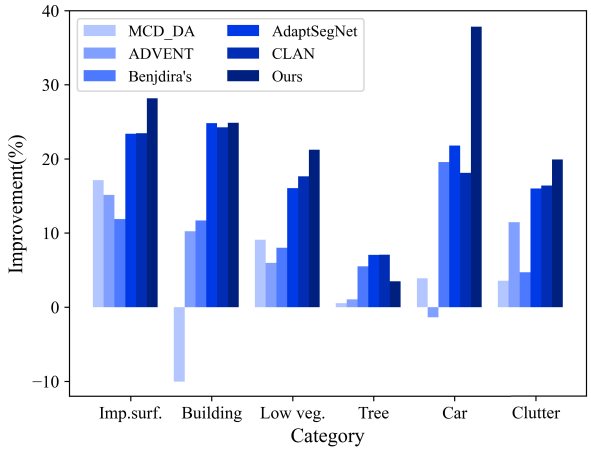

Fig. 9. Improvement F_1 score of each class on comparison methods and our method compared with the baseline Deeplab-v2 (ResNet-101) only trained on source images without adaptation on Vaihingen dataset.

the segmentation model with the translated dataset. The image-to-image translation can only make the source domain images similar to the target domain images in terms of image style. Thus, the difference in scene context and category representation is difficult to reduce. The testing results of this method are less effective on objects with large category representation differences, such as tree and car, as shown in Fig. 10(c). AdaptSegNet and CLAN can achieve more precise predictions due to the output space alignment. However, their performance in categories, such as car and impervious surfaces, is still unsatisfactory, as shown in Fig. 10(d) and (e). By considering the local semantic consistency and paying the attention to the unaligned regions and categories, as shown in Fig. 10(f), our method achieves the best and clearest testing results. Specifically, the classification effects of cars, impervious surfaces, and clutter are more accurate and smoother than those obtained using the comparison methods.

Table II shows the number of parameters and floating point operations (FLOPs) of all tested methods. We can find that the two-stage method "Benjdira's" has the largest number of parameters and FLOPs compared with the other one-stage methods. This is because it first needs a GAN to transform the images of the source domain into the target domain, and then train the semantic segmentation network on the translated source domain images. Among all other one-stage methods, due to adding a CCA module to the basic adversarial framework, our method spends the biggest parameters and FLOPs. Although our method took more 10.87M parameters compared to AdaptSegNet and more $0.16 * 10^{11}$ FLOPs compared to CLAN, its mIoU on the Potsdam $\rightarrow$ Vaihingen task has increased by 3%–17% compared to the other methods, which makes the calculation increments meaningful.

2) Comparative Studies on Potsdam $\rightarrow$ Gaofen-2: To further validate the effectiveness of the proposed algorithm, we also conduct comparative experiments on the task Potsdam (RGB) $\rightarrow$ Gaofen-2 (RGB) with the state-of-the-art domain adaptation methods [28], [29], [31], [34], [39] listed in Table III. In this table, the proposed method achieves the highest OA, MA, and mIoU values of 60.59%, 47.99%, and

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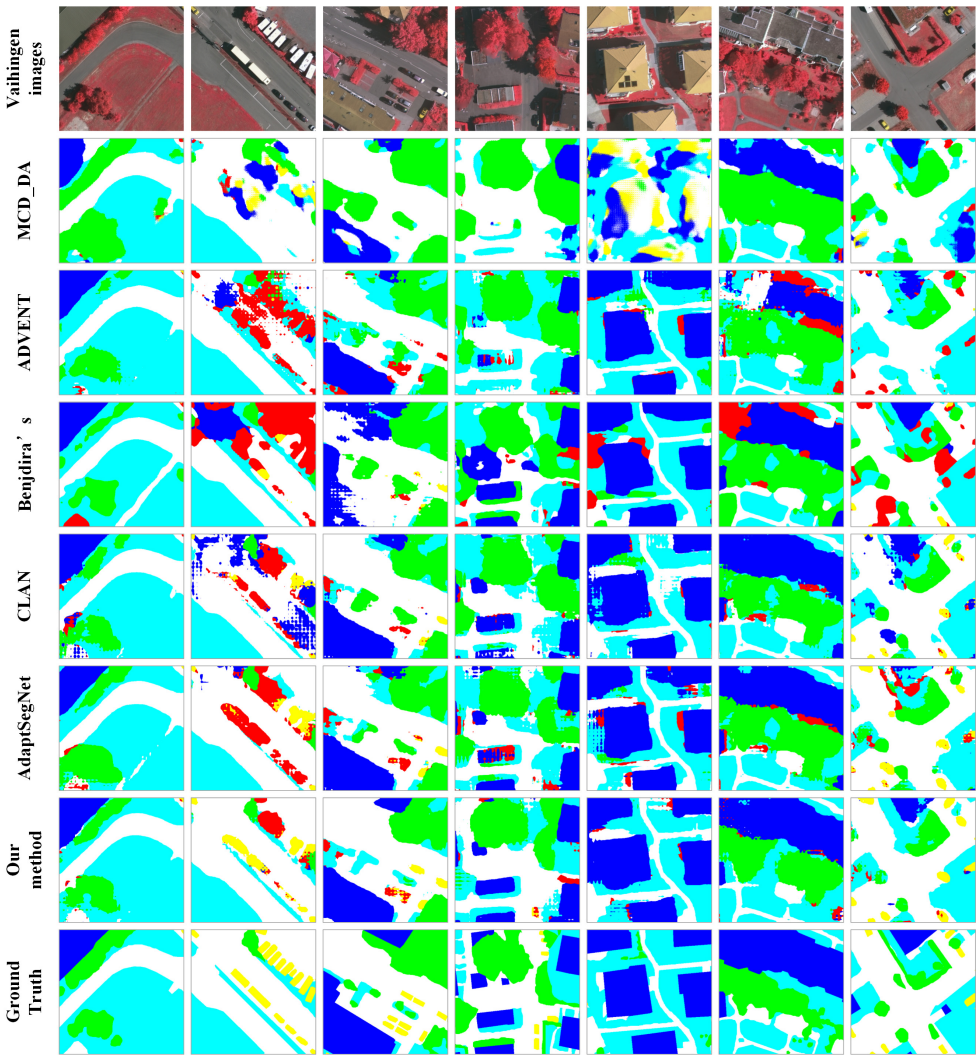

Fig. 10. Qualitative analysis results on Vaihingen dataset.

TABLE II
NUMBER OF PARAMETERS AND FLOPS OF ALL TESTED METHODS

Methods	MCD_DA [28]	ADVENT [29]	Benjdira's [39]	AdaptSegNet [31]	CLAN [34]	Ours
Parameters	99.3M	186M	272.8M	186M	166M	196.87M
FLOPs	$1.11*10^{11}$	$1.86*10^{11}$	$5.72*10^{11}$	$1.86*10^{11}$	$1.93*10^{11}$	$2.09*10^{11}$

34.02% on the task Potsdam → Gaofen-2, respectively, compared with other state-of-the-art methods, which demonstrate that the proposed method can also achieve excellent cross-domain semantic segmentation performance on Potsdam → Gaofen-2. However, this highest score is significantly lower than that on the task of Potsdam → Vaihingen, which indicates that the domain difference of from Potsdam → gaofen-2 is greater than that of from Potsdam → Vaihingen.

From Table III, we can also see that the F_1 score of our method in the categories of Impervious Surfaces, Building, Low Vegetation, Car, and Clutter reaches the best accuracies of 74.17%, 51.38%, 55.78%, 15.88%, and 25.15%, respectively, compared with those of other models. Meanwhile, Fig. 11 shows that our method has a great improvement in F_1 score in each category except for Tree category compared with the baseline trained on source images without adaptation, which indicates that the strategy of adding attention to the unaligned categories and regions can improve the cross-domain classification performance on categories. The qualitative analysis results from Potsdam → Gaofen-2 are shown in Fig. 12. Similarly, our method performs the best compared to other methods, especially for the categories of impervious surface, building, and clutter.

3) Ablation Study: We present some ablation studies on the main components of the proposed model. Table IV shows the results of baseline model Deeplab-v2 (ResNet-101), which is only trained on the source dataset without adaptation. This method is compared with our adapted models under

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TABLE III
RESULT EVALUATION OF COMPARATIVE EXPERIMENTS ON THE ADAPTATION TASK POTSDAM → GAOFEN-2

Methods	Imp. surf.	Building	Low veg.	Tree	Car	Clutter	OA	MA	mIoU
Baseline	47.30	42.14	52.43	76.90	2.05	11.73	52.76	39.94	27.15
MCD_DA [28]	37.97	31.24	54.78	73.29	9.83	2.30	50.92	40.04	24.13
ADVENT [29]	67.93	46.79	55.06	**80.45**	8.92	9.46	60.36	44.36	32.81
Benjdira's [39]	67.57	44.47	52.68	65.14	5.50	12.34	51.56	40.23	27.56
AdaptSegNet [31]	68.34	45.58	50.97	67.02	9.34	20.52	57.48	44.37	30.68
CLAN [34]	67.71	46.38	45.31	69.29	6.59	9.29	57.18	41.50	28.66
Ours	**74.17**	**51.38**	**55.78**	65.71	**15.88**	**25.15**	**60.59**	**47.99**	**34.02**

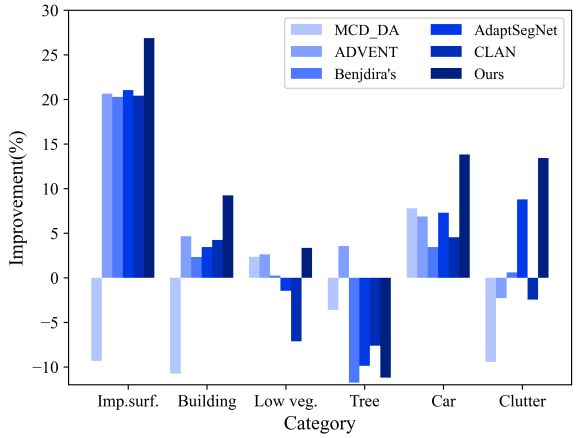

Fig. 11. Improvement F_1 score of each class on comparison methods and our method compared with the baseline Deeplab-v2 (ResNet-101) only trained on source images without adaptation on Gaofen-2 dataset.

different settings, including feature adaptation with global-level discriminator (aligning the feature space by adversarial learning, feature + G_D), adding category-level discriminator (feature + G_D & C_D), and adding category-certainty attention module (feature + G_D & C_D + CCA). Compared with the baseline model, employing feature adaptation with a global-level discriminator yields a result of 67.33%, 54.40%, and 38.53% in OA, MA, and mIoU, which brings 18.48%, 6.22%, and 12.60% improvement. This finding emphasizes the importance of feature space alignment for cross-domain semantic segmentation. Meanwhile, employing the global & category-level discriminator based on the feature adaptation during training outperforms the feature adaptation model only aligning the feature space using global-level discriminator by 4.88%, 5.74%, and 4.84% in OA, MA, and mIoU. Moreover, this method solves the problem of negative transfer on Tree category caused by feature adaptation (F_1 score of feature adaptation is reduced by 3.26% compared with the baseline model without domain adaptation, as shown in Table IV and Fig. 13). This performance is caused by ignoring the local category semantic consistency during the global alignment of the feature adaptation method. Furthermore, when we add the category-certainty attention module on the basis of adapted model using global & category-level discriminator, the improvement segmentation performance shows values of 1.41%, 2.89%, and 2.56% in OA, MA, and mIoU. Fig. 13 shows that, compared with the model using global & category-level discriminator, adding the category-certainty attention module can greatly improve the segmentation performance on the categories that are hard to adapt (such as low vegetation, tree, and clutter). Therefore, category-certainty attention module can greatly benefit the cross-domain semantic segmentation of HRSIs.

Fig. 14 presents the Over Accuracy, Mean Accuracy, and mIoU curve of our adapted methods on the Vaihingen validation dataset during training. As shown in this figure, the curve of our adapted method with feature adaptation tends to decline sharply after approximately 7000 iterations and then remains stable after 40 000 iterations. However, a downward trend is still observed. This result shows that the feature adaptation method lacks local semantic consistency while pursuing global alignment, which leads to negative transfer. On the contrary, the curves of our adapted methods with global & category-level adversarial discriminator and category-certainty attention module can maintain a steady upward trend after 7000 iterations. This capability shows that our method can consider the consistency of local semantics while pursuing global domain adaptation.

4) Visual Analysis: Fig. 15 shows the importance of category-level certainty saliency map created by global & category-level discriminator in cross domain semantic segmentation of HRSIs. The range of the saliency map is [0, 1], where 0 presents the source domain, 1 presents the target domain. The bigger the value of the saliency map, the more possible that the discriminator judges the input region from the target domain, indicating this region has not been well adapted. On the contrary, the small value of the saliency map indicates the discriminator can accept the region from the source domain indicating a well domain adaption. Take the Building as an example, the yellow box in Fig. 15 points out a saliency map with big value, showing a not well adaption. As a result, Fig. 15(b) exhibits a bad performance in semantic segmentation. After inputting the saliency map to the CCA module, the classifier can selectively pay attention to the regions that are aligned or misaligned according to the category-certainty of the saliency map. Consequently, the better performance of cross domain semantic segmentation can be shown in Fig. 15(c).

5) Impact of Resolution Difference: In this section, we discuss the effect of image resolution on domain adaptation. Tables V and VI show the results on the task Potsdam → Vaihingen and the task Potsdam → Gaofen-2 with our proposed method trained on the source and target patches with the same size (512 × 512, 512 × 512) and the scaled size (960 × 960 and 512 × 512), respectively. In Table V,

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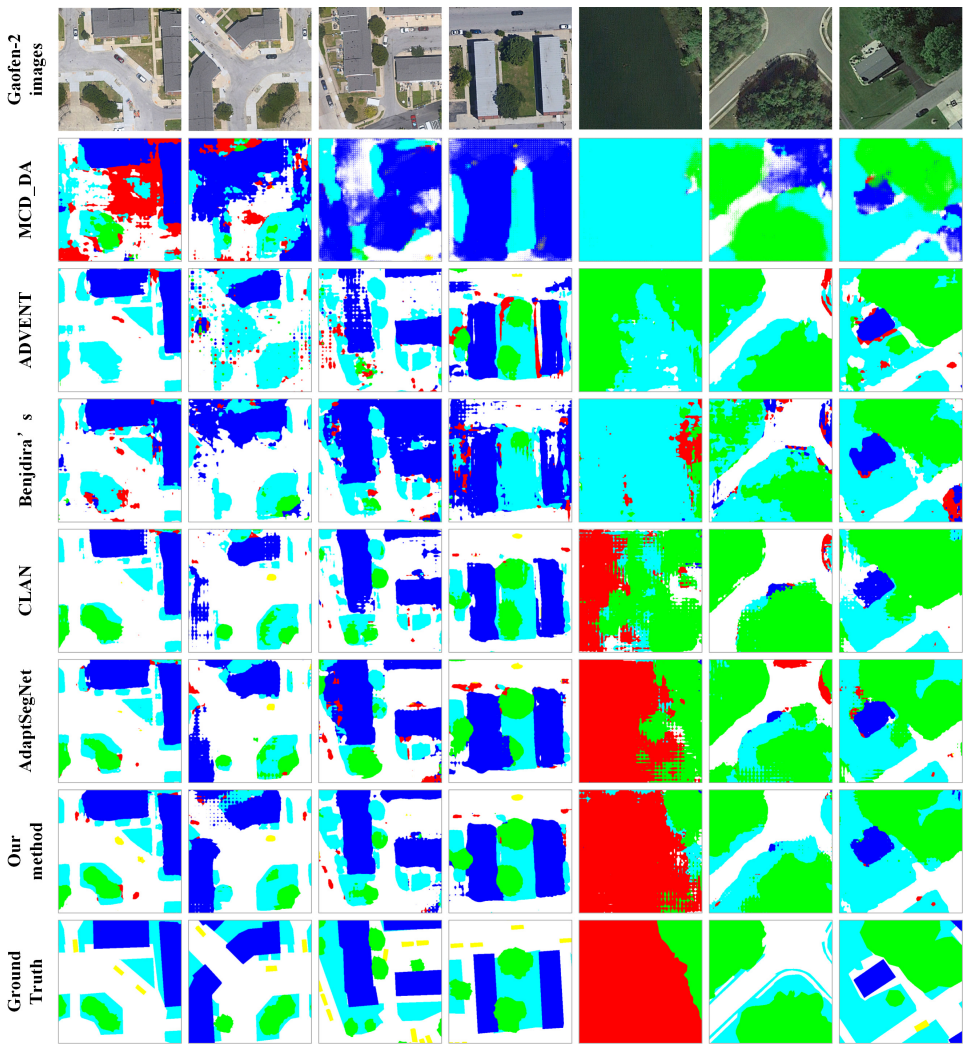

Fig. 12. Qualitative analysis results on Gaofen-2 dataset.

TABLE IV
RESULTS OF THE ABLATION STUDIES

Methods	Imp. surf.	Building	Low veg.	Tree	Car	Clutter	OA	MA	mIoU
Baseline	52.12	59.38	44.36	60.69	2.74	8.92	48.88	48.18	25.93
Ours (feature + G_D)	74.18	74.14	60.95	57.43	30.99	19.58	67.33	54.40	38.53
Ours (feature + G_D & C_D)	79.29	80.10	64.52	61.91	34.82	22.80	72.21	60.14	43.37
Ours (feature + G_D & C_D + CCA)	**80.30**	**84.24**	**65.59**	64.18	**40.57**	**28.85**	**73.62**	**63.03**	**45.91**

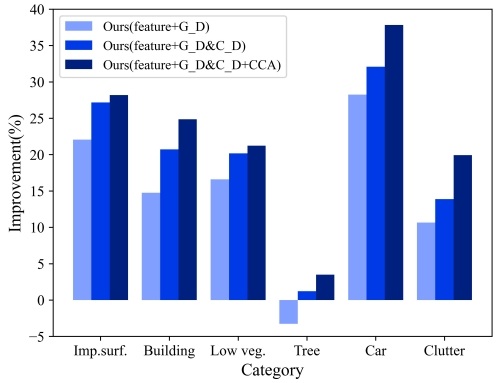

Fig. 13. Improvement F_1 score of each class on the proposed method compared with the baseline Deeplab-v2 (ResNet-101) only trained on source images without adaptation.

compared with the baseline Deeplab-v2 (ResNet-101) without adaptation, our proposed method trained with same size patches on the task Potsdam → Vaihingen yields a result of 66.02%, 58.15%, and 39.39% in OA, MA, and mIoU, which brings 17.14%, 9.97%, and 13.46% improvement. However, when we train the model on the scaled image patches, we obtain the further result of 7.60%, 4.88%, and 6.52% improvement in OA, MA, and mIoU compared with the model trained on the same size patches. In Table VI, on the task Potsdam → Gaofen-2, the model trained on the scaled image patches also obtain the improvement of 3.4%, 1.75%, and 2.18% in OA, MA, and mIoU compared with the model trained on the same size patches. Therefore, keeping the objects of the source and target domains at the same scale can improve the performance of the domain adaptation model.

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TABLE V

RESULTS OF THE EFFECT OF RESOLUTION DIFFERENCE ON THE TASK POTSDAM → VAIHINGEN

Methods	Imp. surf.	Building	Low veg.	Tree	Car	Clutter	OA	MA	mIoU
Baseline	52.12	59.38	44.36	60.69	2.74	8.92	48.88	48.18	25.93
Ours (1:1)	73.22	72.79	58.49	63.37	35.10	21.95	66.02	58.15	39.39
Ours (9:5)	**80.30**	**84.24**	**65.59**	**64.18**	**40.57**	**28.85**	**73.62**	**63.03**	**45.91**

TABLE VI

RESULTS OF THE EFFECT OF RESOLUTION DIFFERENCE ON THE TASK POTSDAM → GAOFEN-2

Methods	Imp. surf.	Building	Low veg.	Tree	Car	Clutter	OA	MA	mIoU
Baseline	47.30	42.14	52.43	**76.90**	2.05	11.73	52.76	39.94	27.15
Ours (1:1)	66.60	48.90	52.92	64.76	14.30	12.73	57.19	46.24	31.84
Ours (9:5)	**74.17**	**51.38**	**55.78**	65.71	**15.88**	**25.15**	**60.59**	**47.99**	**34.02**

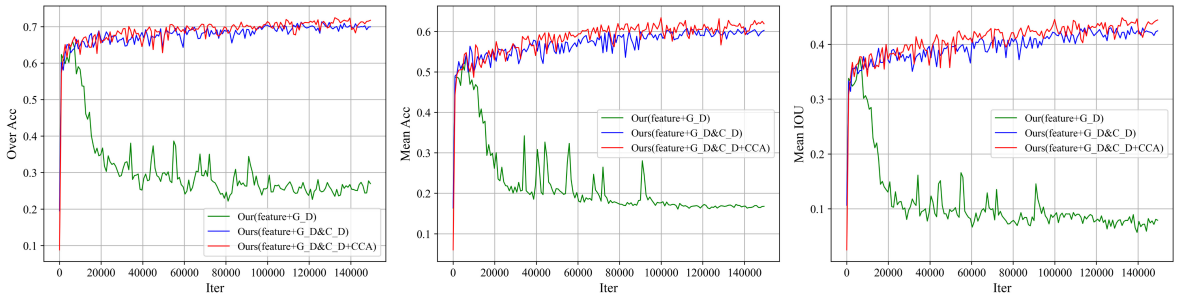

Fig. 14. Over Accuracy, Mean Accuracy, and mIoU curve of our methods on Vaihingen validation dataset during training.

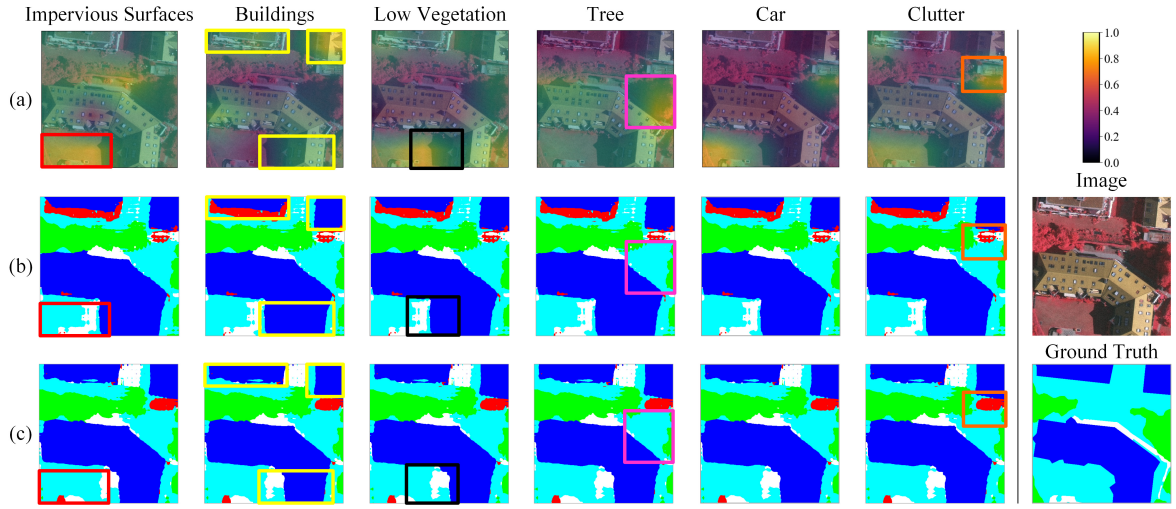

Fig. 15. Visualization results of category-certainty. (a) Saliency map of category-certainty. (b) Ours (feature + G_D & C_D). (c) Ours (feature + G_D & C_D + CCA).

IV. CONCLUSION AND FUTURE WORK

In this study, we explore a novel unsupervised domain adaptation method to handle the cross-domain semantic segmentation in HRSIs. We tackle the domain adaptation problem via adversarial learning in the feature space. We employ a global and category-level adversarial loss to enhance local semantic consistency during the process of global alignment for avoiding negative transfer to further enhance the adapted model. We also design a category-certainty attention module to improve the performance of the classifier on the target domain by focusing the training on the misaligned regions and categories. The experimental results show the effectiveness of the proposed method, which performs favorably against baseline models and state-of-the-art algorithms. Our method also greatly improves the segmentation performance on the categories that are hard to adapt (such as low vegetation, tree, and clutter).

In our experiments, the prior spatial layout can be beneficial in improving the difference between categories. Therefore, we will explore the use of the context prior obtained from label map in the future work to provide rich spatial layout information for cross-domain semantic segmentation of HRSIs.

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REFERENCES

[1] M. J. Swain and D. H. Ballard, "Indexing via color histograms," in *Proc. 3rd Int. Conf. Comput. Vis.*, Dec. 1991, pp. 390–393.

[2] M. Umaselvi, S. S. Kumar, and M. Athithya, "Color based urban and agricultural land classification by GLCM texture features," in *Proc. IET Chennai 3rd Int. Conf. Sustain. Energy Intell. Syst. (SEISCON)*, 2012, pp. 192–195.

[3] S. Qi, J. Ma, J. Lin, Y. Li, and J. Tian, "Unsupervised ship detection based on saliency and S-HOG descriptor from optical satellite images," *IEEE Geosci. Remote Sens. Lett.*, vol. 12, no. 7, pp. 1451–1455, Jul. 2015.

[4] D. G. Lowe, "Distinctive image features from scale-invariant keypoints," *Int. J. Comput. Vis.*, vol. 20, pp. 91–110, Nov. 2003.

[5] V. Vapnik, "Statistical Learning Theory," *Ann. Inst. Stat. Math.*, vol. 55, no. 2, pp. 371–389, 1998.

[6] T. K. Ho, "The random subspace method for constructing decision forests," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 20, no. 8, pp. 832–844, Aug. 1998.

[7] K. Fukushima, S. Miyake, and T. Ito, "Neocognitron: A neural network model for a mechanism of visual pattern recognition," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-13, no. 5, pp. 826–834, Sep. 1983.

[8] E. Shelhamer, J. Long, and T. Darrell, "Fully convolutional networks for semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 4, pp. 640–651, Apr. 2017.

[9] V. Badrinarayanan, A. Kendall, and R. Cipolla, "SegNet: A deep convolutional encoder-decoder architecture for image segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 39, no. 12, pp. 2481–2495, Dec. 2017.

[10] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional networks for biomedical image segmentation," in *Proc. Int. Conf. Med. Image Comput. Comput.-Assist. Intervent.*, 2015, pp. 234–241.

[11] H. Zhao, J. Shi, X. Qi, X. Wang, and J. Jia, "Pyramid scene parsing network," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 6230–6239.

[12] L.-C. Chen, G. Papandreou, I. Kokkinos, K. Murphy, and A. L. Yuille, "DeepLab: Semantic image segmentation with deep convolutional nets, atrous convolution, and fully connected CRFs," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 40, no. 4, pp. 834–848, Apr. 2017.

[13] C. Zhang *et al.*, "An object-based convolutional neural network (OCNN) for urban land use classification," *Remote Sens. Environ.*, vol. 216, pp. 57–70, Oct. 2018.

[14] J. Chen, J. Zhu, G. Sun, J. Li, and M. Deng, "SMAF-net: Sharing multiscale adversarial feature for high-resolution remote sensing imagery semantic segmentation," *IEEE Geosci. Remote Sens. Lett.*, vol. 18, no. 11, pp. 1921–1925, Nov. 2021, doi: 10.1109/LGRS.2020.3011151.

[15] J. Chen, F. He, Y. Zhang, G. Sun, and M. Deng, "SPMF-Net: Weakly supervised building segmentation by combining superpixel pooling and multi-scale feature fusion," *Remote Sens.*, vol. 12, no. 6, p. 1049, Mar. 2020.

[16] V. M. Patel, R. Gopalan, R. Li, and R. Chellappa, "Visual domain adaptation: A survey of recent advances," *IEEE Signal Process. Mag.*, vol. 32, no. 3, pp. 53–69, May 2015.

[17] Y. Ganin and V. Lempitsky, "Unsupervised domain adaptation by backpropagation," 2014, *arXiv:1409.7495*.

[18] Y. Ganin *et al.*, "Domain-adversarial training of neural networks," *J. Mach. Learn. Res.*, vol. 17, no. 1, pp. 2030–2096, 2017.

[19] M. Long, Y. Cao, J. Wang, and M. I. Jordan, "Learning transferable features with deep adaptation networks," 2015, *arXiv:1502.02791*.

[20] M. Long, H. Zhu, J. Wang, and M. I. Jordan, "Unsupervised domain adaptation with residual transfer networks," in *Proc. Neural Inf. Process. Syst.*, 2016, pp. 136–144.

[21] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell, "Adversarial discriminative domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2962–2971.

[22] D. Sejdinovic, B. Sriperumbudur, A. Gretton, and K. Fukumizu, "Equivalence of distance-based and RKHS-based statistics in hypothesis testing," *Ann. Statist.*, vol. 41, no. 5, pp. 2263–2291, 2013.

[23] I. J. Goodfellow *et al.*, "Generative adversarial nets," in *Proc. Int. Conf. Neural Inf. Process. Syst.*, 2014, pp. 2672–2680.

[24] J. Shen, Y. Qu, W. Zhang, and Y. Yu, "Wasserstein distance guided representation learning for domain adaptation," 2017, *arXiv:1707.01217*.

[25] Z. Pei, Z. Cao, M. Long, and J. Wang, "Multi-adversarial domain adaptation," in *Proc. Nat. Conf. Artif. Intell.*, 2018, pp. 3934–3941.

[26] J. Hoffman, D. Wang, F. Yu, and T. Darrell, "FCNs in the wild: Pixel-level adversarial and constraint-based adaptation," 2016, *arXiv:1612.02649*.

[27] Y. Zhang, P. David, and B. Gong, "Curriculum domain adaptation for semantic segmentation of urban scenes," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2020–2030.

[28] K. Saito, K. Watanabe, Y. Ushiku, and T. Harada, "Maximum classifier discrepancy for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3723–3732.

[29] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Pérez, "ADVENT: Adversarial entropy minimization for domain adaptation in semantic segmentation," 2018, *arXiv:1811.12833*.

[30] Y.-H. Chen, W.-Y. Chen, Y.-T. Chen, B.-C. Tsai, Y.-C.-F. Wang, and M. Sun, "No more discrimination: Cross city adaptation of road scene segmenters," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2011–2020.

[31] Y.-H. Tsai, W.-C. Hung, S. Schulter, K. Sohn, M.-H. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7472–7481.

[32] L. Zhang *et al.*, "Unsupervised domain adaptation using robust class-wise matching," *IEEE Trans. Circuits Syst. Video Technol.*, vol. 29, no. 5, pp. 1339–1349, May 2019.

[33] Y. Pan, T. Yao, Y. Li, Y. Wang, C.-W. Ngo, and T. Mei, "Transferrable prototypical networks for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2239–2247.

[34] Y. Luo, L. Zheng, T. Guan, J. Yu, and Y. Yang, "Taking a closer look at domain shift: Category-level adversaries for semantics consistent domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2507–2516.

[35] F. S. Saleh, M. S. Aliakbarian, M. Salzmann, L. Petersson, and J. M. Alvarez, "Effective use of synthetic data for urban scene semantic segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 84–100.

[36] J. Hoffman *et al.*, "CyCADA: Cycle-consistent adversarial domain adaptation," 2017, *arXiv:1711.03213*.

[37] Z. Wu *et al.*, "DCAN: Dual channel-wise alignment networks for unsupervised scene adaptation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 518–534.

[38] Y. Li, L. Yuan, and N. Vasconcelos, "Bidirectional learning for domain adaptation of semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6936–6945.

[39] B. Benjdira, Y. Bazi, A. Koubaa, and K. Ouni, "Unsupervised domain adaptation using generative adversarial networks for semantic segmentation of aerial images," *Remote Sens.*, vol. 11, no. 11, p. 1369, 2019.

[40] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.

[41] B. Fang, R. Kou, L. Pan, and P. Chen, "Category-sensitive domain adaptation for land cover mapping in aerial scenes," *Remote Sens.*, vol. 11, no. 22, p. 2631, Nov. 2019.

[42] A. L. Maas, A. Y. Hannun, and A. Y. Ng, "Rectifier nonlinearities improve neural network acoustic models," in *Proc. ICML*, vol. 30, no. 1, 2013.

[43] D. Bahdanau, K. Cho, and Y. Bengio, "Neural machine translation by jointly learning to align and translate," 2014, *arXiv:1409.0473*.

[44] J. Chen, L. Wan, J. Zhu, G. Xu, and M. Deng, "Multi-scale spatial and channel-wise attention for improving object detection in remote sensing imagery," *IEEE Geoscience Remote Sens. Lett.*, vol. 17, no. 4, pp. 681–685, Apr. 2020.

[45] K. Xu *et al.*, "Show, attend and tell: Neural image caption generation with visual attention," in *Proc. Int. Conf. Mach. Learn.*, PMLR, 2015, pp. 2048–2057.

[46] J. Chen, Y. Han, L. Wan, X. Zhou, and M. Deng, "Geospatial relation captioning for high-spatial-resolution images by using an attention-based neural network," *Int. J. Remote Sens.*, vol. 40, no. 16, pp. 6482–6498, Aug. 2019.

[47] W. Chan, N. Jaitly, Q. Le, and O. Vinyals, "Listen, attend and spell: A neural network for large vocabulary conversational speech recognition," in *Proc. IEEE Int. Conf. Acoust., Speech Signal Process. (ICASSP)*, Mar. 2016, pp. 4960–4964.

[48] G. Kang, L. Zheng, Y. Yan, and Y. Yang, "Deep adversarial attention alignment for unsupervised domain adaptation: The benefit of target expectation maximization," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, 2018, pp. 420–436.

[49] V. K. Kurmi, S. Kumar, and V. P. Namboodiri, "Attending to discriminative certainty for domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 491–500.

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[50] *ISPRS Potsdam 2D Semantic Labeling Dataset*. Accessed: Dec. 10, 2017. [Online]. Available: http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-potsdam.html
[51] *ISPRS Vaihingen 2D Semantic Labeling Dataset*. Accessed: Dec. 10, 2017. [Online]. Available: http://www2.isprs.org/commissions/comm3/wg4/2d-sem-label-vaihingen.html
[52] *High Resolution Visible Image Fine Annotation Dataset*. Accessed: 2019. [Online]. Available: http://sw.chreos.org/dataset/3
[53] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," 2014, *arXiv:1412.6980*.

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